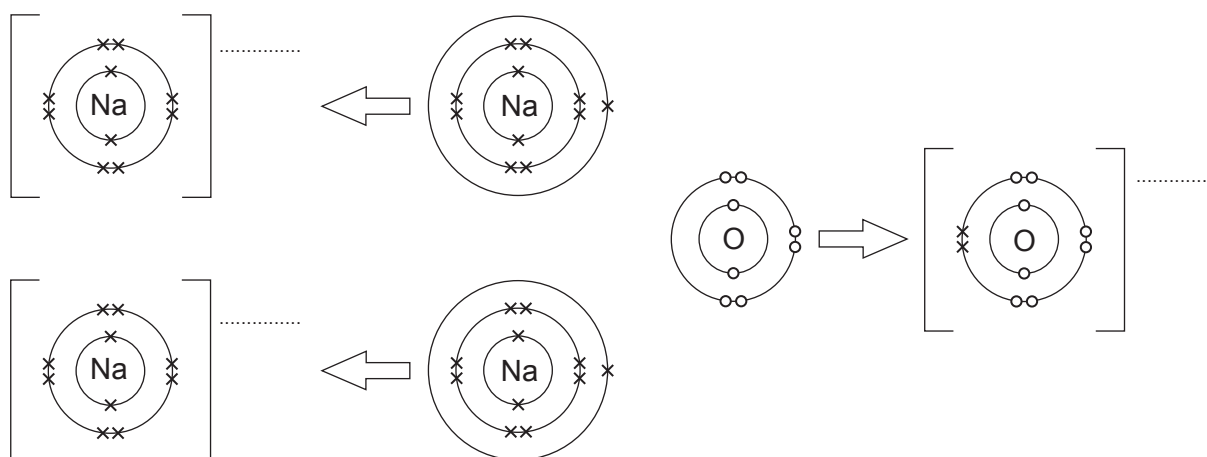


2. (a) The table gives the electronic structures of sodium and oxygen.

Element	Electronic structure
sodium	2,8,1
oxygen	2,6

- (i) Sodium reacts with oxygen to give sodium oxide.

The diagram below can be used to show the electronic changes that occur as sodium oxide is formed.



Complete the diagram by

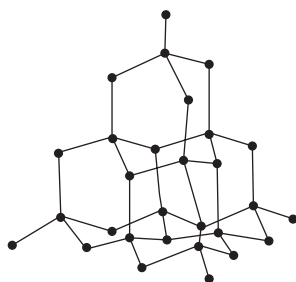
- **drawing arrows** to show the movement of electrons between the sodium and oxygen atoms,
- giving the **charges** of the sodium and oxide ions that are formed.

[2]

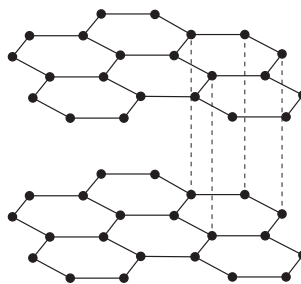


- (ii) Draw a dot and cross diagram to show the bonding in an oxygen molecule, O_2 . [2]

- (b) (i) Diamond and graphite are both made from carbon atoms. Their structures are shown below.



diamond



graphite

- I. State which structure is able to conduct electricity. Give a reason for your answer. [1]

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.....

- II. State which structure is used as a lubricant. Give a reason for your answer. [1]

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- (ii) Give the name of another form of carbon that is used in drug delivery systems. [1]

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